

智能手机应用程序开发 作业

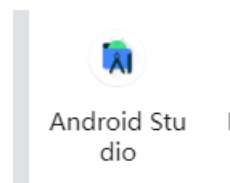
计算机 **2101**

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第一章

我安装的程序为



运行虚拟机



第二章

P73 2.2 实现多选题的界面效果。

p73 2.3 实现选择出生日期的界面效果

```
public class MainActivity extends Activity {
    TextView mTextView;           //TextView 对象，用于显示问题
    CheckBox mCheckBox1;          //CheckBox 对象，用于显示选项 1
    CheckBox mCheckBox2;          //CheckBox 对象，用于显示选项 2
    CheckBox mCheckBox3;          //CheckBox 对象，用于显示选项 3
    CheckBox mCheckBox4;          //CheckBox 对象，用于显示选项 4
    Button mButton;               //Button 对象，用于显示提交按钮
    int checkedcount = 0;         //计数器，用于统计选中的个数
    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); //加载布局文件
        //加载控件
        mTextView = (TextView)this.findViewById(R.id.textview);
        mCheckBox1 = (CheckBox)this.findViewById(R.id.checkbox1);
        mCheckBox2 = (CheckBox)this.findViewById(R.id.checkbox2);
        mCheckBox3 = (CheckBox)this.findViewById(R.id.checkbox3);
```

```

mCheckBox4 = (CheckBox)this.findViewById(R.id.checkbox4);
mButton = (Button)this.findViewById(R.id.button);
//选项 1 事件监听
mCheckBox1.setOnCheckedChangeListener(new
OnCheckedChangeListener() {
    public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked) {
        if (mCheckBox1.isChecked()) {
            checkedcount++;
            DisplayToast("你选择了：" + mCheckBox1.getText());
        } else {
            checkedcount--; } } });
//选项 2 事件监听
mCheckBox2.setOnCheckedChangeListener(new
OnCheckedChangeListener() {
    public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked) {
        if (mCheckBox2.isChecked()) {
            checkedcount++;
            DisplayToast("你选择了：" + mCheckBox2.getText());
        } else {
            checkedcount--;
        } } } });
//选项 3 事件监听
mCheckBox3.setOnCheckedChangeListener(new
OnCheckedChangeListener() {
    public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked) {
        if (mCheckBox3.isChecked()) {
            checkedcount++;
            DisplayToast("你选择了：" + mCheckBox3.getText());
        } else {
            checkedcount--;
        } } } });
//选项 4 事件监听
mCheckBox4.setOnCheckedChangeListener(new
OnCheckedChangeListener() {
    public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked) {
        if (mCheckBox4.isChecked()) {
            checkedcount++;
            DisplayToast("你选择了：" + mCheckBox4.getText());
        } else {
            checkedcount--;

```

```

        } } });

    ///提交“按钮事件监听
    mButton.setOnClickListener(new OnClickListener() {
        public void onClick(View v) {
            DisplayToast("谢谢参与，你一共选择了" + checkedcount + "项！");
        } });
    public void DisplayToast(String string) {
        Toast mToast = Toast.makeText(this, string, Toast.LENGTH_LONG);
        mToast.setGravity(Gravity.TOP, 0, 450);
        mToast.show();
    }
    class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final DatePicker datePicker = findViewById(R.id.date_picker);
        datePicker.updateDate(2021, 0, 1);
        datePicker.setOnDateChangeListener((view, year, monthOfYear,
        dayOfMonth) ->
        {
            Calendar calendar = Calendar.getInstance();
            calendar.set(year, monthOfYear, dayOfMonth);
            @SuppressWarnings("SimpleDateFormat") SimpleDateFormat simpleDateFormat =
            new SimpleDateFormat("yyyy 年 MM 月 dd 日");
            Toast.makeText(getApplicationContext(), "你的出身日期为：" +
            simpleDateFormat.format(calendar.getTime()), Toast.LENGTH_SHORT).show();
        });
        Button button2 = findViewById(R.id.button2);
        button2.setOnClickListener(v -> {
            Calendar calendar = Calendar.getInstance();
            calendar.set(datePicker.getYear(), datePicker.getMonth(),
            datePicker.getDayOfMonth());
            @SuppressWarnings("SimpleDateFormat") SimpleDateFormat simpleDateFormat =

```

```

new SimpleDateFormat("yyyy 年 MM 月 dd 日");
Toast.makeText(getApplicationContext(), " 你的 出生 日期 为 ： " +
simpleDateFormat.format(calendar.getTime()), Toast.LENGTH_SHORT).show();
}); }

```



第三章

p87 3.1 实现注册界面的布局效果，至少包括用户名、密码、忘记密码、登录等控件。

```

public class MainActivity extends AppCompatActivity {
    @Override

```

```

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);} }

```

```

<TableLayout xmlns:android="
http://schemas.android.com/apk/res/android"
    android:layout_width="300dp"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:collapseColumns="2"
    android:stretchColumns="1">

```

```

<!--第 1 行-->

```

```

<TableRow>

```

```

<!--第 1 列-->

```

```

<TextView

```

```

    android:layout_width="wrap_content"

```

```

    android:layout_height="wrap_content"

```

```

    android:text="用户名" />

```

```

<!--第 2 列-->

```

```

<EditText

```

```

    android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName" />
    </TableRow>
    <!--第 2 行-->
    <TableRow>
        <!--第 1 列-->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="密码" />
        <!--第 2 列-->
        <EditText
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:ems="10"android:inputType="textPassword" />
        <!--第 3 列-->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="忘记密码-+" />
    </TableRow>
    <!--第 3 行-->
    <TableRow>
        <!--第 2 列-->
        <TextView
            android:id="@+id/textView4"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_column="1"
            android:gravity="end"
            android:text="注册账号" />
    </TableRow>
    <!--第 4 行-->
    <TableRow>
        <!--第 1 列和第 2 列合并-->
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_span="2" android:text="登录" />
    </TableRow>
</TableLayout>

```



第四章：

p122 4.3 实现京东首页的轮播广告效果，至少包括 3 个产品广告的图像。

```
public class MainActivity extends AppCompatActivity implements  
View.OnClickListener {
```

```
    private ViewPager mViewPager;  
    private TextView mPagePositionTextView;  
    private TextView mPageStateTextView;  
    private ImageView mPage0;  
    private ImageView mPage1;  
    private ImageView mPage2;  
    private ImageView mPage3;  
    private int mCurrentPosition;  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
        mPagePositionTextView = findViewById(R.id.text_view_page_position);  
        mPageStateTextView = findViewById(R.id.text_view_page_state);  
        mPage0 = findViewById(R.id.image_view_page0);  
        mPage1 = findViewById(R.id.image_view_page1);  
        mPage2 = findViewById(R.id.image_view_page2);  
        mPage3 = findViewById(R.id.image_view_page3);  
        mPage0.setOnClickListener(this);  
        mPage1.setOnClickListener(this);  
        mPage2.setOnClickListener(this);  
        mPage3.setOnClickListener(this);  
        ImageView leftImageView = findViewById(R.id.image_button_left);  
        ImageView rightImageView = findViewById(R.id.image_button_right);  
        leftImageView.setOnClickListener(this);  
        rightImageView.setOnClickListener(this);  
        mViewPager = findViewById(R.id.view_pager);  
        mViewPager.addOnPageChangeListener(new PageChangeListener());  
        init();    }  
    // 初始化
```

```

private void init(){
    ArrayList<View> pagerViews = new ArrayList<>();
    pagerViews.add(View.inflate(this, R.layout.view_pager_welcome0, null));
    pagerViews.add(View.inflate(this, R.layout.view_pager_welcome1, null));
    pagerViews.add(View.inflate(this, R.layout.view_pager_welcome2, null));
    pagerViews.add(View.inflate(this, R.layout.view_pager_welcome3, null));
    WelcomePagerAdapter welcomePagerAdapter = new
WelcomePagerAdapter(pagerViews);
    mViewPager.setAdapter(welcomePagerAdapter); }
//单击监听事件
@Override
public void onClick(View v) {
    int id = v.getId();
    if (id == R.id.image_button_left) {
        if (mCurrentPosition > 0) {
            mCurrentPosition--;}
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } else if (id == R.id.image_button_right) {
        if (mCurrentPosition < 3) {
            mCurrentPosition++; }
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } else if (id == R.id.image_view_page0) {
        mCurrentPosition = 0;
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } else if (id == R.id.image_view_page1) {
        mCurrentPosition = 1;
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } else if (id == R.id.image_view_page2) {
        mCurrentPosition = 2;
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } else if (id == R.id.image_view_page3) {
        mCurrentPosition = 3;
        mViewPager.setCurrentItem(mCurrentPosition, true);
    } }
//页面改变监听器
public class PageChangeListener implements
ViewPager.OnPageChangeListener {
    //页面选择事件
    public void onPageSelected(int position) {
        //翻页时当前 page，改变当前状态圆点图片。
        mCurrentPosition = position;
        switch (position) {
            case 0:
                mPage0.setImageResource(R.mipmap.img_page_now);

```



```

        mPage1.setImageResource(R.mipmap.img_page);
        mPage2.setImageResource(R.mipmap.img_page);
        mPage3.setImageResource(R.mipmap.img_page);
        break;
    case 1:
        mPage1.setImageResource(R.mipmap.img_page_now);
        mPage0.setImageResource(R.mipmap.img_page);
        mPage2.setImageResource(R.mipmap.img_page);
        mPage3.setImageResource(R.mipmap.img_page);
        break;
    case 2:
        mPage2.setImageResource(R.mipmap.img_page_now);
        mPage0.setImageResource(R.mipmap.img_page);
        mPage1.setImageResource(R.mipmap.img_page);
        mPage3.setImageResource(R.mipmap.img_page);
        break;
    case 3:
        mPage3.setImageResource(R.mipmap.img_page_now);
        mPage0.setImageResource(R.mipmap.img_page);
        mPage1.setImageResource(R.mipmap.img_page);
        mPage2.setImageResource(R.mipmap.img_page);
        break;    }}

//页面滚动事件
@SuppressLint("SetTextI18n")
public void onPageScrolled(int position, float positionOffset, int
positionOffsetPixels) {
    //mPagePositionTextView.setText("页面索引号:" + position + " 偏移百分
比:" + positionOffset + " 偏移像素:" + positionOffsetPixels); }
//页面滚动状态改变事件
public void onPageScrollStateChanged(int state) {
    switch (state) {
        case ViewPager.SCROLL_STATE_IDLE:
            mPageStateTextView.setText("空闲状态");
            break;
        case ViewPager.SCROLL_STATE_DRAGGING:
            mPageStateTextView.setText("拖动状态");
            break;
        case ViewPager.SCROLL_STATE_SETTLING:
            mPageStateTextView.setText("结束状态");
            break;
    } } }

```



第五章

p150 5.1 实现登录后显示用户名的效果。输入用户名、密码后登录，启动一个新的 Activity

显示用户名。

由于注册界面的代码不是本次作业的要求，故不展示

登录界面后端业务逻辑代码：

```
public class MainActivity extends AppCompatActivity {
    private EditText edtUsername;
    private EditText edtPassword;
    private Button btnLogin;
    private Button btnCancel;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        edtUsername=findViewById(R.id.et_user_name);
        edtPassword=findViewById(R.id.et_psw);
        Button bt2=findViewById(R.id.btn_login);
        Button bt1=findViewById(R.id.tv_register);
        bt1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent =new Intent(MainActivity.this,RegisterActivity.class);
                startActivity(intent);
            } });
        bt2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String strUsername = edtUsername.getText().toString().trim();
                String strPassword = edtPassword.getText().toString().trim();
                if (strUsername.equals("admin") && strPassword.equals("admin")) {
```

```

        Toast.makeText(MainActivity.this,
            "用户名和密码正确!", Toast.LENGTH_SHORT).show();
        Intent intent =new Intent(MainActivity.this,MainActivity2.class);
        intent.putExtra("Username",strUsername);
        intent.putExtra("Password",strPassword);
        startActivity(intent);
    } else {
        Toast.makeText(MainActivity.this,
            "用户名或密码错误请重试!", Toast.LENGTH_SHORT).show();} }
}); }

```

个人主页后端业务代码：

```

public class MainActivity2 extends AppCompatActivity {
    private TextView tvMessage;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
        tvMessage=findViewById(R.id.et_username);
        Intent intent =getIntent();
        if(intent!=null){
            String username = intent.getStringExtra("Username");
            String password = intent.getStringExtra("Password");
            String message = "登录成功!\n 用户:" + username + "\n 密码:" +
password;
            tvMessage.setText(message);
        }
        Button btn=findViewById(R.id.btn_quit);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                finish();}    }); }
}

```

登陆界面前端代码：

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/login_bg"
    android:orientation="vertical">
    <!--标题栏-->
    <include layout="@layout/main_title_bar"></include>
    <!--显示头像，记得加入 id iv_head -->
    <ImageView
        android:id="@+id/iv_head"
        android:layout_width="70dp"

```

```
        android:layout_height="70dp"
        android:layout_marginTop="25dp"
        android:layout_gravity="center_horizontal"
        android:background="@drawable/default_icon"/>
<!--输入框-->
<EditText
    android:id="@+id/et_user_name"
    android:layout_width="fill_parent"
    android:layout_height="48dp"
    android:layout_marginTop="35dp"
    android:layout_marginLeft="35dp"
    android:layout_marginRight="35dp"
    android:layout_gravity="center_horizontal"
    android:drawableLeft="@drawable/user_name_icon"
    android:drawablePadding="10dp"
    android:paddingLeft="8dp"
    android:gravity="center_vertical"
    android:hint="请输入用户名"
    android:singleLine="true"
    android:textColor="#000000"
    android:textColorHint="#a3a3a3"
    android:textSize="14sp"/>
<!--输入框-->
<EditText
    android:id="@+id/et_psw"
    android:layout_width="fill_parent"
    android:layout_height="48dp"
    android:layout_gravity="center_horizontal"
    android:layout_marginLeft="35dp"
    android:layout_marginRight="35dp"
    android:drawableLeft="@drawable/psw_icon"
    android:drawablePadding="10dp"
    android:paddingLeft="8dp"
    android:gravity="center_vertical"
    android:hint="请输入密码"
    android:inputType="textPassword"
    android:singleLine="true"
    android:textColor="#000000"
    android:textColorHint="#a3a3a3"
    android:textSize="14sp"/>
<!--按钮-->
<Button
    android:id="@+id/btn_login"
    android:layout_width="fill_parent"
```

```

        android:layout_height="40dp"
        android:layout_marginTop="15dp"
        android:layout_marginLeft="35dp"
        android:layout_marginRight="35dp"
        android:layout_gravity="center_horizontal"
        android:background="@drawable/register_selector"
        android:text="登 录"
        android:textColor="@android:color/white"
        android:textSize="18sp"/>
<!--显示 tv register , find_psw -->
<LinearLayout
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:layout_marginTop="8dp"
    android:layout_marginLeft="35dp"
    android:layout_marginRight="35dp"
    android:gravity="center_horizontal"
    android:orientation="horizontal">
    <Button
        android:id="@+id/tv_register"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:gravity="center_horizontal"
        android:padding="8dp"
        android:text="立即注册"
        android:textColor="@android:color/white"
        android:textSize="14sp" />
    <!--layout_weight="1" layout_width="0dp"实现均分效果-->
    <Button
        android:id="@+id/tv_find_psw"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:gravity="center_horizontal"
        android:padding="8dp"
        android:text="找回密码?"
        android:textColor="@android:color/white"
        android:textSize="14sp" />
    </LinearLayout>
</LinearLayout>

```

输入用户名

输入密码

登录

第六章

p176 6.3 通过广播接收器实现两个 App 之间通过自定义广播进行数据通信。

```
public class MyReceiver1 extends BroadcastReceiver {
    private static final String TAG = "MyBroadcastReceiver";
    @Override
    public void onReceive(Context context, Intent intent) {
        Log.d(TAG, "-----Receivers1-----");
        Log.d(TAG, "Action: " + intent.getAction());
        Log.d(TAG, "URI: " + intent.toUri(Intent.URI_INTENT_SCHEME));
        Log.d(TAG, "自定义广播的 info : " + intent.getStringExtra("info"));
    }
}
```

AppCompatActivity.java 代码：

```
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        //发送标准广播
        findViewById(R.id.button).setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v) {
                Intent intent=new Intent();
                intent.setAction("MyBroadcastReceiver.Custom");
                intent.putExtra("info","显示广播");
                intent.addFlags(Intent.FLAG_ACTIVITY_PREVIOUS_IS_TOP);
                sendBroadcast(intent);
            }
        });
    }
}
```

```

        }    }); }
public class MyReceiver2 extends BroadcastReceiver {
    private static final String TAG = "MyBroadcastReceiver";
    @Override
    public void onReceive(Context context, Intent intent) {
        // TODO: This method is called when the BroadcastReceiver is receiving
        // an Intent broadcast.
        Log.d(TAG, "-----Receiver-----");
        Log.d(TAG, "Action: " + intent.getAction());
        Log.d(TAG, "URI: " + intent.toUri(Intent.URI_INTENT_SCHEME));
        // 自定义广播
        if ("MyBroadcastReceiver.Custom".equals(intent.getAction())) {
            Log.d(TAG, "自定义广播的 info : " + intent.getStringExtra("info"));
        }
    }
}

AppCompatActivity.java
public class MainActivity extends AppCompatActivity {
    private MyReceiver2 myReceiver = new MyReceiver2();
    private boolean mRegistered = false;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); //注册广播接收器
        IntentFilter intentFilter = new IntentFilter();
        intentFilter.addAction(ConnectivityManager.CONNECTIVITY_ACTION);
        intentFilter.addAction(Intent.ACTION_SCREEN_ON);
        intentFilter.addAction(Intent.ACTION_SCREEN_OFF);
        intentFilter.addAction(Intent.ACTION_USER_PRESENT);
        intentFilter.addAction(Intent.ACTION_BATTERY_OKAY);
        intentFilter.addAction(Intent.ACTION_BATTERY_LOW);
        intentFilter.addAction(Intent.ACTION_BATTERY_CHANGED);
        //          intentFilter.addAction("com.vt.c0701.MyBroadcastReceiver.ss");
        registerReceiver(myReceiver, intentFilter);
        mRegistered = true;
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        if(mRegistered) {
            unregisterReceiver(myReceiver);
        }
    }
}

```



第七章：

p213 7.2 使用 SQLite 实现记录登录时间的功能，并且能够查询登录的历史记录。

```
public class HistoryActivity extends AppCompatActivity
implements View.OnClickListener {
    private Button cancel;
    public EditText mIdEditText;
    public EditText mNameEditText;
    public EditText mPhoneEditText;
    private Context mContext;
    private SQLiteDatabase mSQLiteDatabase;
    private ContactSQLiteOpenHelper
mContactSQLiteOpenHelper;
    private String mDataBaseName = "contacts.db";
    private String mTableName = "contacts";
    private List<ContactModel> mContactModel;
    private ContactListAdapter mContactListAdapter;
    private ListView mContactListView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        setContentView(R.layout.activity_history);
        super.onCreate(savedInstanceState);
        this.setTitle("登录历史");
        mContext = this;
        mContactModel = new ArrayList<>();
        mContactListView =
        findViewById(R.id.list_view_contacts);
```



```

//findViewById(R.id.btn_list).setOnClickListener(this);
init();
cancel = (Button) findViewById(R.id.btn_cancel);
findViewById(R.id.btn_cancel).setOnClickListener(new
View.OnClickListener(){
public void onClick(View view) {
finish();
}
});
}
// 初始化
private void init(){
mContactSQLiteOpenHelper = new
ContactSQLiteOpenHelper(mContext, mDataBaseName, null, 1);
mSQLiteDataBase =
mContactSQLiteOpenHelper.getWritableDatabase();
mContactListAdapter = new ContactListAdapter(this,
mContactModel);
mContactListView.setAdapter(mContactListAdapter);
listAllContacts();
}
// 显示所有联系人
private void listAllContacts() {
mContactModel.clear();
Cursor cursor = mSQLiteDataBase.rawQuery("SELECT *
FROM " + mTableName + " order by id asc", null);
while (cursor.moveToNext()) { String id =
cursor.getString(cursor.getColumnIndex("id"));
String name =
cursor.getString(cursor.getColumnIndex("name"));
String phone =
cursor.getString(cursor.getColumnIndex("phone"));
mContactModel.add(new ContactModel(id, name, phone));
}
cursor.close();
// 更新 ListView
mContactListAdapter.notifyDataSetChanged();
}
@Override
public void onClick(View v) {
String id = mIdEditText.getText().toString();
String name = mNameEditText.getText().toString();
String phone = mPhoneEditText.getText().toString();
switch (v.getId()) { } }

```

```

public class ContactListAdapter extends BaseAdapter {
    private HistoryActivity mContext;
    private List<ContactModel> mContactModel;
    private ViewHolder mViewHolder;
    //构造方法 public ContactListAdapter(HistoryActivity mContext,
    List<ContactModel> mContactModel) {
        this.mContext = mContext;
        this.mContactModel = mContactModel;}
    @Override
    public int getCount() {
        return mContactModel.size(); }
    @Override
    public Object getItem(int position) {
        return mContactModel;}
    @Override
    public long getItemId(int position) {
        return position;}
    //获取列表项视图
    @Override
    public View getView(int position, View convertView,
    ViewGroup parent) {
        if (convertView == null) {//convertView 未实例化时
            convertView =
            LayoutInflater.from(mContext).inflate(R.layout.list_view_
            item_contact, parent, false);
            mViewHolder = new ViewHolder();
            mViewHolder.idTextView =
            convertView.findViewById(R.id.text_view_id);
            mViewHolder.nameTextView =
            convertView.findViewById(R.id.text_view_name);
            mViewHolder.phoneTextView =
            convertView.findViewById(R.id.text_view_phone);
            convertView.setTag(mViewHolder);//holder 缓存到
            convertView
        } else {//convertView 实例化时
            mViewHolder = (ViewHolder)
            convertView.getTag();//从 convertView 获取 holder 缓存
            mViewHolder.idTextView.setText(mContactModel.get(position)
            .id);
            mViewHolder.idTextView.setId(position);
            mViewHolder.idTextView.setOnClickListener(mTextViewOnClic
            kListener); mViewHolder.nameTextView.setText(mContactModel.get(positi
            on).name);
            mViewHolder.nameTextView.setId(position);

```

```

mViewHolder.nameTextView.setOnClickListener(mTextViewOnClick
ClickListener);
mViewHolder.phoneTextView.setText(mContactModel.get(posit
ion).phone);
mViewHolder.phoneTextView.setId(position);
mViewHolder.phoneTextView.setOnClickListener(mTextViewOnC
lickListener);
return convertView; }
private View.OnClickListener
mTextViewOnClickListener = new View.OnClickListener() {
public void onClick(View v)
{ (mContext.mIdEditText).setText(mContactModel.get(v.
getId()).id);
(mContext.mNameEditText).setText(mContactModel.get(v.getI
d()).name);
(mContext.mPhoneEditText).setText(mContactModel.get(v.get
Id()).phone);}};
class ViewHolder {
TextView idTextView;
TextView nameTextView;
TextView phoneTextView;}}
public class ContactSQLiteOpenHelper extends
SQLiteOpenHelper {
public ContactSQLiteOpenHelper(Context context, String
name, SQLiteDatabase.CursorFactory factory, int version){
super(context, name, factory, version);}
@Override
public void onCreate(SQLiteDatabase db) {
// 创建 contacts 表
db.execSQL("CREATE TABLE contacts(id INTEGER
PRIMARY KEY AUTOINCREMENT,name VARCHAR(16),phone
VARCHAR(11))");}
@Override
public void onUpgrade(SQLiteDatabase db, int
oldVersion, int newVersion) {
if (newVersion > oldVersion) { // contacts 表添加 notes 列
db.execSQL("ALTER TABLE contacts ADD notes
VARCHAR(20)"); }}

```



第八章：

p271 8.1 使用 Camera2 实现定时拍照的功能。

首先自定义拍照会用到 SurfaceView 控件显示照片的预览区域，以下是布局文件：

```
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent" >
    <SurfaceView
        android:id="@+id/surface_view"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical" >
        <ImageView
            android:id="@+id/start"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_centerHorizontal="true"
            android:layout_alignParentBottom="true"
            android:layout_marginBottom="10dp"
            android:src="@drawable/capture"/>
        <TextView
            android:id="@+id/count_down"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:layout_gravity="center"
            android:gravity="center"
```

```

        android:textSize="80sp"/>
    </RelativeLayout>
</FrameLayout>
MainActivity.java
public class MainActivity extends AppCompatActivity implements
SurfaceHolder.Callback,
    View.OnClickListener, Camera.PictureCallback {
    private SurfaceView mSurfaceView;
    private ImageView mlvStart;
    private TextView mTvCountDown;
    private SurfaceHolder mHolder;
    private Camera mCamera;
    private Handler mHandler = new Handler();
    private int mCurrentTimer = 10;
    private boolean mIsSurfaceCreated = false;
    private boolean mIsTimerRunning = false;
    private static final int CAMERA_ID = 0; //后置摄像头
// private static final int CAMERA_ID = 1; //前置摄像头
    private static final String TAG = MainActivity.class.getSimpleName();
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        initView();
        initEvent(); }
    @Override
    protected void onPause() {
        super.onPause();
        stopPreview(); }
    private void initView() {
        mSurfaceView = (SurfaceView) findViewById(R.id.surface_view);
        mlvStart = (ImageView) findViewById(R.id.start);
        mTvCountDown = (TextView) findViewById(R.id.count_down); }
    private void initEvent() {
        mHolder = mSurfaceView.getHolder();
        mHolder.addCallback(this);
        mlvStart.setOnClickListener(this); }
    @Override
    public void surfaceCreated(SurfaceHolder holder) {
        mIsSurfaceCreated = true; }
    @Override
    public void surfaceChanged(SurfaceHolder holder, int format, int width, int
height) {
        startPreview(); }

```

```

@Override
public void surfaceDestroyed(SurfaceHolder holder) {
    mIsSurfaceCreated = false; }
private void startPreview() {
    if (mCamera != null || !mIsSurfaceCreated) {
        Log.d(TAG, "startPreview will return");
        return; }
    mCamera = Camera.open(CAMERA_ID);
    Camera.Parameters parameters = mCamera.getParameters();
    int width = getResources().getDisplayMetrics().widthPixels;
    int height = getResources().getDisplayMetrics().heightPixels;
    Camera.Size size = getBestPreviewSize(width, height, parameters);
    if (size != null) {
        //设置预览分辨率
        parameters.setPreviewSize(size.width, size.height);
        //设置保存图片的大小
        parameters.setPictureSize(size.width, size.height); }
    //自动对焦
    parameters.setFocusMode(Camera.Parameters.FOCUS_MODE_AUTO);
    parameters.setPreviewFrameRate(20);
    //设置相机预览方向
    mCamera.setDisplayOrientation(90);
    mCamera.setParameters(parameters);
    try {
        mCamera.setPreviewDisplay(mHolder);
    } catch (Exception e) {
        Log.d(TAG, e.getMessage()); }
    mCamera.startPreview(); }
private void stopPreview() {
    //释放 Camera 对象
    if (mCamera != null) {
        try {
            mCamera.setPreviewDisplay(null);
            mCamera.stopPreview();
            mCamera.release();
            mCamera = null;
        } catch (Exception e) {
            Log.e(TAG, e.getMessage());
        } } }
private Camera.Size getBestPreviewSize(int width, int height,
    Camera.Parameters parameters) {
    Camera.Size result = null;
    for (Camera.Size size : parameters.getSupportedPreviewSizes()) {
        if (size.width <= width && size.height <= height) {

```

```

        if (result == null) {
            result = size;
        } else {
            int resultArea = result.width * result.height;
            int newArea = size.width * size.height;
            if (newArea > resultArea) {
                result = size; } } } return result; }

@Override
public void onClick(View v) {
    switch (v.getId()) {
        case R.id.start:
            if (!mIsTimerRunning) {
                mIsTimerRunning = true;
                mHandler.post(timerRunnable); }
            break; } }

private Runnable timerRunnable = new Runnable() {
    @Override
    public void run() {
        if (mCurrentTimer > 0) {
            mTvCountDown.setText(mCurrentTimer + "");
            mCurrentTimer--;
            mHandler.postDelayed(timerRunnable, 1000);
        } else {
            mTvCountDown.setText("");
            mCamera.takePicture(null, null, null, MainActivity.this);
            playSound();
            mIsTimerRunning = false;
            mCurrentTimer = 10; } } };

@Override
public void onPictureTaken(byte[] data, Camera camera) {
    try {
        FileOutputStream fos = new FileOutputStream(new File
            (Environment.getExternalStorageDirectory() + File.separator +
                System.currentTimeMillis() + ".png")); //旋转角度，保证保存的图片方向是对的
        Bitmap bitmap = BitmapFactory.decodeByteArray(data, 0, data.length);
        Matrix matrix = new Matrix();
        matrix.setRotate(90);
        bitmap = Bitmap.createBitmap(bitmap, 0, 0,
            bitmap.getWidth(), bitmap.getHeight(), matrix, true);
        bitmap.compress(Bitmap.CompressFormat.PNG, 100, fos);
        fos.flush();
        fos.close();
    } catch (FileNotFoundException e) {
        e.printStackTrace();
    }
}

```

```

    } catch (IOException e) {
        e.printStackTrace(); }
    mCamera.startPreview(); }
/**
 * 播放系统拍照声音
 */
public void playSound() {
    MediaPlayer mediaPlayer = null;
    AudioManager audioManager = (AudioManager)
getSystemService(Context.AUDIO_SERVICE);
    int volume =
audioManager.getStreamVolume( AudioManager.STREAM_NOTIFICATION);
    if (volume != 0) {
        if (mediaPlayer == null)
            mediaPlayer = MediaPlayer.create(this,
                Uri.parse("file:///system/media/audio/ui/camera_click.ogg"));
        if (mediaPlayer != null) {
            mediaPlayer.start();
        } } } }

```

